

# Linear Programming: Formulations

## Example 1:

A manufacturer has two resources  $R_1$  and  $R_2$  and is making two products  $P_1$  and  $P_2$ . Each unit of  $P_1$  requires 1 unit of  $R_1$  and 4 units of  $R_2$ . Each unit of  $P_2$  requires 2 units of  $R_1$  and 1 unit of  $R_2$ . The manufacturer has 8 units of  $R_1$  and 14 units of  $R_2$ . The profit for  $P_1$  is 4\$ per unit and for  $P_2$  it is 7\$ per unit. Find how many units of both products should be produced to get the maximum profit.

## Example 2:

A calculator company produces a scientific calculator and a graphing calculator. Long-term projections indicate an expected demand of at least 100 scientific and 80 graphing calculators each day. Because of limitations on production capacity, no more than 200 scientific and 170 graphing calculators can be made daily. To satisfy a shipping contract, a total of at least 200 calculators must be shipped each day.

If each scientific calculator sold results in a \$2 loss, but each graphing calculator produces a \$5 profit, how many of each type should be made daily to maximize net profits?

# Linear Programming: Formulations (do it yourself)

1. A farmer has 10 acres to plant in wheat and rye. He has to plant at least 7 acres. However, he has only \$1200 to spend and each acre of wheat costs \$200 to plant and each acre of rye costs \$100 to plant. Moreover, the farmer has to get the planting done in 12 hours and it takes an hour to plant an acre of wheat and 2 hours to plant an acre of rye. If the profit is \$500 per acre of wheat and \$300 per acre of rye how many acres of each should be planted to maximize profits?
2. A gold processor has two sources of gold ore, source A and source B. In order to keep his plant running, at least three tons of ore must be processed each day. Ore from source A costs \$20 per ton to process, and ore from source B costs \$10 per ton to process. Costs must be kept to less than \$80 per day. Moreover, Federal Regulations require that the amount of ore from source B cannot exceed twice the amount of ore from source A. If ore from source A yields 2 oz. of gold per ton, and ore from source B yields 3 oz. of gold per ton, how many tons of ore from both sources must be processed each day to maximize the amount of gold extracted subject to the above constraints?
3. A publisher has orders for 600 copies of a certain text from San Francisco and 400 copies from Sacramento. The company has 700 copies in a warehouse in Novato and 800 copies in a warehouse in Lodi. It costs \$5 to ship a text from Novato to San Francisco, but it costs \$10 to ship it to Sacramento. It costs \$15 to ship a text from Lodi to San Francisco, but it costs \$4 to ship it from Lodi to Sacramento. How many copies should the company ship from each warehouse to San Francisco and Sacramento to fill the order at the least cost?

# Linear Programming: Simplex Method

**Maximize  $z: 6X_1 + 5X_2$**

**Such that,**

$$X_1 + X_2 \leq 5$$

$$3X_1 + 2X_2 \leq 12$$

**Add slack variables ( $s_1$  and  $s_2$ ) to convert inequality into equality,**

$$X_1 + X_2 + s_1 = 5$$

$$3X_1 + 2X_2 + s_2 = 12$$

# Simplex Method: Theoretical Basis

## Iteration 1

First Basic feasible solution is obtained by getting the maximum feasible values for slack variables

$$S_1 = 5 - X_1 - X_2 \quad \text{-----(1)}$$

$$S_2 = 12 - 3X_1 - 2X_2 \quad \text{-----(2)}$$

And  $z = 6X_1 + 5X_2$

Here basic feasible solution is  $z = 0$  for  $X_1 = 0$ ,  $X_2 = 0$ ,  $S_1 = 5$ ,  $S_2 = 12$

## Iteration 2:

We observe the variable with highest coefficient in  $z$ , that variable is  $X_1$ , we then try to increase  $z$  by increasing  $X_1$ , **the maximum feasible value of  $X_1$**  is 4 from eqn (2), increasing  $X_1$  beyond 4 will make  $s_1$  negative thus violating the constraint of all the variables being positive, therefore,

$$X_1 = 4 - \frac{2}{3}X_2 - \frac{1}{3}s_2 \quad \text{----(3)} \quad // \text{ substituting for } X_1 \text{ in (2)}$$

and  $z$

$$S_1 = 5 - X_1 - X_2 = 1 - \frac{1}{3}X_2 + \frac{1}{3}s_2 \quad \text{-----(4)}$$

$$\text{And } z = 24 + X_2 - 2s_2$$

# Simplex Method: Theoretical Basis

## Iteration 3

We observe the variable with highest positive coefficient in  $z$ , that variable is  $X_2$ , we then try to increase  $z$  by increasing this variable, **the maximum feasible value of  $X_2$**  is 3 from eqn (4), increasing  $X_1$  beyond 3 will make  $s_2$  negative thus violating the constraint of all the variables being positive, therefore,

$$X_2 = 3 + s_2 - 3s_1$$

$$X_1 = 2 + 2s_1 - s_2$$

// substituting in (3) and  $z$

$$\text{And } z = 27 - 3s_1 - s_2$$

Notice that  $z$  can not be increased further because the coefficients of all the variables ( $s_1$  and  $s_2$ ) are negative and increasing them will decrease the value of  $z$ . Therefore, the final solution is  $z = 27$  for  $X_2 = 3$  and  $X_1 = 2$  and value of slack variables are zero.

# Simplex Method: Tabular Form

**Problem 1: Maximize  $z: 6X_1+5x_2$**

**s.t.**

$$x_1+x_2 \leq 5$$

$$3x_1+2x_2 \leq 12$$

$$x_1, x_2 \geq 0$$

**Introduce slack variables to convert it into equality,**

$$x_1+x_2 +s_1 = 5$$

$$3x_1+2x_2 +s_2 =12$$

$$x_1, x_2, s_1, s_2 \geq 0$$

**In objective function, contribution of  $s_1$  and  $s_2$  are 0.**

**The first basic feasible solution is found by setting original variables ( $X_1$  and  $X_2$ ) to 0. We start with  $s_1$  and  $s_2$ .**

# Simplex Method: Problem 1 contd.

Corresponding contribution of variables in Z, objective function

$0$   $s_1$   
 $0$   $s_2$

$C_j$     6    5    0    0  
 $X_1$      $X_2$      $s_1$      $s_2$     RHS    L (leaving)

|             |          |          |          |          |           |  |
|-------------|----------|----------|----------|----------|-----------|--|
|             | <b>1</b> | <b>1</b> | <b>1</b> | <b>0</b> | <b>5</b>  |  |
|             | <b>3</b> | <b>2</b> | <b>0</b> | <b>1</b> | <b>12</b> |  |
| $Z_j$       | 0        | 0        | 0        | 0        | 0         |  |
| $C_j - Z_j$ | 6        | 5        | 0        | 0        | 0         |  |
|             |          |          |          |          |           |  |
|             |          |          |          |          |           |  |
|             |          |          |          |          |           |  |
|             |          |          |          |          |           |  |
|             |          |          |          |          |           |  |

$Z_j$  values are computed by multiplying corresponding contribution of variables in objective function with corresponding coefficients of variables in the equation, such as,  $0 \cdot 1 + 0 \cdot 3 = 0$ , and so on.

$C_j - Z_j$  is computed by subtracting  $Z_j$  from corresponding  $C_j$  values.

# Simplex Method: Problem 1 contd.

Corresponding contribution of variables in Z, objective function

|         | $C_j$       | 6     | 5     | 0     | 0     |     |             |
|---------|-------------|-------|-------|-------|-------|-----|-------------|
|         |             | $X_1$ | $X_2$ | $s_1$ | $s_2$ | RHS | L (leaving) |
| 0 $s_1$ |             | 1     | 1     | 1     | 0     | 5   | 5/1 = 5     |
| 0 $s_2$ |             | 3     | 2     | 0     | 1     | 12  | 12/3 = 4 →  |
| $s_1$   | $Z_j$       | 0     | 0     | 0     | 0     | 0   |             |
|         | $C_j - Z_j$ | 6     | 5     | 0     | 0     | 0   |             |
| $s_2$   |             |       |       |       |       |     |             |
|         |             |       |       |       |       |     |             |
|         |             |       |       |       |       |     |             |
|         |             |       |       |       |       |     |             |
|         |             |       |       |       |       |     |             |

Entering variable is the one with highest positive  $C_j - Z_j$  values, in this step it is  $X_1$ . The leaving variable is the one with lowest corresponding (RHS/corresponding coefficient) of entering variable, in this case 1 and 3 are those coefficients. Thus  $s_2$  leaves. Corresponding element, 3 (circled in the table) in this case is the pivot element and Row  $R_2$  is pivot Row.

# Simplex Method: Problem 1 contd.

Corresponding contribution of variables in Z, objective function

|   |       | $C_j$       | 6     | 5     | 0     | 0     |     |             |       |
|---|-------|-------------|-------|-------|-------|-------|-----|-------------|-------|
|   |       |             | $X_1$ | $X_2$ | $s_1$ | $s_2$ | RHS | L (leaving) |       |
| 0 | $s_1$ |             | 1     | 1     | 1     | 0     | 5   | $5/1 = 1$   | $R_1$ |
| 0 | $s_2$ |             | 3     | 2     | 0     | 1     | 12  | $12/3 = 4$  | $R_2$ |
|   |       | $Z_j$       | 0     | 0     | 0     | 0     | 0   |             |       |
|   |       | $C_j - Z_j$ | 6     | 5     | 0     | 0     |     |             |       |
| 0 | $s_1$ |             | 0     | 1/3   | 1     | -1/3  |     |             | $R_3$ |
| 6 | $X_1$ |             | 1     | 2/3   | 0     | 1/3   |     |             | $R_4$ |
|   |       |             |       |       |       |       |     |             |       |
|   |       |             |       |       |       |       |     |             |       |
|   |       |             |       |       |       |       |     |             |       |
|   |       |             |       |       |       |       |     |             |       |
|   |       |             |       |       |       |       |     |             |       |
|   |       |             |       |       |       |       |     |             |       |

$X_1$  enters the basis and  $s_2$  leaves. We divide the  $R_2$  by pivot element to get the value corresponding to  $X_1$  1

Next, Now we have to get a 0 above the corresponding 1 in the last step, to achieve that, we subtract  $R_4$  From  $R_1$  to get  $R_3$

# Simplex Method: Problem 1 contd.

|   |             | $C_j$ | 6     | 5     | 0     | 0      |     |             |       |
|---|-------------|-------|-------|-------|-------|--------|-----|-------------|-------|
|   |             |       | $X_1$ | $X_2$ | $s_1$ | $s_2$  | RHS | L (leaving) |       |
| 0 | $s_1$       |       | 1     | 1     | 1     | 0      | 5   | $5/1 = 1$   | $R_1$ |
| 0 | $s_2$       |       | 3     | 2     | 0     | 1      | 12  | $12/3 = 4$  | $R_2$ |
|   | $Z_j$       |       | 0     | 0     | 0     | 0      | 0   |             |       |
|   | $C_j - Z_j$ |       | 6     | 5     | 0     | 0      |     |             |       |
| 0 | $s_1$       |       | 0     | $1/3$ | 1     | $-1/3$ | 1   | $1/1/3 = 3$ | $R_3$ |
| 6 | $X_1$       |       | 1     | $2/3$ | 0     | $1/3$  | 4   | $4/2/3 = 6$ | $R_4$ |
|   | $Z_j$       |       | 6     | 4     | 0     | 2      | 24  |             |       |
|   | $C_j - Z_j$ |       | 0     | 1     | 0     | -2     |     |             |       |
|   |             |       |       |       |       |        |     |             |       |
|   |             |       |       |       |       |        |     |             |       |

We compute corresponding  $Z_j$  and  $C_j - Z_j$  values. Like in the last step we identify entering variable as  $X_2$  since corresponding  $C_j - Z_j$  value is highest positive among all. To identify leaving variable we divide corresponding RHS value by corresponding column coefficient values of entering variable. So variable  $X_2$  enters and variable  $s_1$  leaves. Pivot element becomes  $1/3$  and pivot row is  $R_3$

# Simplex Method: Problem 1 contd.

|       |       | $C_j$       | 6        | 5        | 0        | 0        |           |                 |
|-------|-------|-------------|----------|----------|----------|----------|-----------|-----------------|
|       |       |             | $X_1$    | $X_2$    | $s_1$    | $s_2$    | RHS       | L (leaving)     |
| 0     | $s_1$ |             | <b>1</b> | <b>1</b> | <b>1</b> | <b>0</b> | <b>5</b>  | <b>5/1 = 1</b>  |
| 0     | $s_2$ |             | <b>3</b> | <b>2</b> | <b>0</b> | <b>1</b> | <b>12</b> | <b>12/3 = 4</b> |
|       |       | $Z_j$       | 0        | 0        | 0        | 0        | 0         |                 |
|       |       | $C_j - Z_j$ | 6        | 5        | 0        | 0        |           |                 |
| <hr/> |       |             |          |          |          |          |           |                 |
| 0     | $s_1$ |             | 0        | 1/3      | 1        | -1/3     | 1         | 1/1/3=3         |
| 6     | $X_1$ |             | 1        | 2/3      | 0        | 1/3      | 4         | 4/2/3 = 6       |
|       |       | $Z_j$       | 6        | 4        | 0        | 2        | 24        |                 |
|       |       | $C_j - Z_j$ | 0        | 1        | 0        | -2       |           |                 |
| <hr/> |       |             |          |          |          |          |           |                 |
| 5     | $X_2$ |             | 0        | <b>1</b> | 3        | -1       | 3         |                 |
| 6     | $X_1$ |             | 1        | 0        | -2       | 1        | 2         |                 |
|       |       | $Z_j$       | 6        | 5        | 3        | 1        | 27        |                 |
|       |       | $C_j - Z_j$ | 0        | 0        | -3       | -1       |           |                 |

We get value 1 corresponding to entering variable  $X_2$  by dividing the pivot row by pivot element. Therefore,  $R_5 = R_3 / (1/3)$   
 Now we need 0 above the corresponding  $X_2$  values, to get  $R_6$ , therefore,  $R_6 = R_4 - (2/3)R_5$ . Next we compute  $Z_j$  and  $C_j - Z_j$

# Simplex Method: Problem 1 contd.

|       |       | $C_j$       | 6        | 5        | 0        | 0        |           |                 |                      |
|-------|-------|-------------|----------|----------|----------|----------|-----------|-----------------|----------------------|
|       |       |             | $X_1$    | $X_2$    | $s_1$    | $s_2$    | RHS       | L (leaving)     |                      |
| 0     | $s_1$ |             | <b>1</b> | <b>1</b> | <b>1</b> | <b>0</b> | <b>5</b>  | <b>5/1 = 1</b>  | <b>R<sub>1</sub></b> |
| 0     | $s_2$ |             | <b>3</b> | <b>2</b> | <b>0</b> | <b>1</b> | <b>12</b> | <b>12/3 = 4</b> | <b>R<sub>2</sub></b> |
|       |       | $Z_j$       | 0        | 0        | 0        | 0        | 0         |                 |                      |
|       |       | $C_j - Z_j$ | 6        | 5        | 0        | 0        |           |                 |                      |
| <hr/> |       |             |          |          |          |          |           |                 |                      |
| 0     | $s_1$ |             | 0        | 1/3      | 1        | -1/3     | 1         | 1/1/3=3         | <b>R<sub>3</sub></b> |
| 6     | $X_1$ |             | 1        | 2/3      | 0        | 1/3      | 4         | 4/2/3 = 6       | <b>R<sub>4</sub></b> |
|       |       | $Z_j$       | 6        | 4        | 0        | 2        | 24        |                 |                      |
|       |       | $C_j - Z_j$ | 0        | 1        | 0        | -2       |           |                 |                      |
| <hr/> |       |             |          |          |          |          |           |                 |                      |
| 5     | $X_2$ |             | 0        | 1        | 3        | -1       | 3         |                 | <b>R<sub>5</sub></b> |
| 6     | $X_1$ |             | 1        | 0        | -2       | 1        | 2         |                 | <b>R<sub>6</sub></b> |
|       |       | $Z_j$       | 6        | 5        | 3        | 1        | 27        |                 |                      |
|       |       | $C_j - Z_j$ | 0        | 0        | -3       | -1       |           |                 |                      |

All  $C_j - Z_j$  values are either zero or negative, therefore, we found our solution. The maximum Z is the final  $Z_j$  value, in this case 27 and corresponding values for  $X_2$  and  $X_1$  are corresponding RHS values, in this case they are 2 and 3.

# Simplex Method: Tabular Form (Problem 2)

**Problem 1: Minimize  $z: 3x_1+4x_2$**

**s.t.**

$$2x_1+x_2 \geq 5$$

$$x_1+3x_2 \geq 12$$

**Introduce slack variables to convert it into equality,**

$$x_1+x_2 - s_1 = 5$$

$$3x_1+2x_2 - s_2 = 12$$

**We need to convert this problem into maximization problem.**

**Recall that minimization of  $z$  is same as maximization of  $-z$  (negative  $z$ ). Therefore, our problem becomes,**

**Maximize  $z: -3x_1- 4x_2$**

**s.t.**

$$2x_1+x_2 \geq 5$$

$$x_1+3x_2 \geq 12$$

# Simplex Method: Problem 2 contd..

Now we introduce slack variables to convert inequality into equality.

$$2x_1 + x_2 - s_1 = 5$$

$$x_1 + 3x_2 - s_2 = 12$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Notice the coefficient of slack variable is negative because of greater than inequality.

To get the first basic feasible solution, we set original variables to 0 but in this case the resulting values of slack variables  $s_1 = -5$  and  $s_2 = -12$  are negative violating the last constraint. To avoid that we introduce artificial variables,  $a_1, a_2$ .

$$2x_1 + x_2 - s_1 + a_1 = 5$$

$$x_1 + 3x_2 - s_2 + a_2 = 12$$

$$x_1, x_2, s_1, s_2, a_1, a_2 \geq 0$$

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# Simplex Method: Problem 2 contd...

Now we introduce slack variables to convert inequality into equality.

$$2x_1 + x_2 - s_1 = 5$$

$$x_1 + 3x_2 - s_2 = 12$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Notice the coefficient of slack variable is negative because of greater than inequality.

To get the first basic feasible solution, we set original variables to 0 but in this case the resulting values of slack variables  $s_1 = -5$  and  $s_2 = -12$  are negative violating the last constraint. To avoid that we introduce artificial variables,  $a_1, a_2$ .

$$2x_1 + x_2 - s_1 + a_1 = 5$$

$$x_1 + 3x_2 - s_2 + a_2 = 12$$

$$x_1, x_2, s_1, s_2, a_1, a_2 \geq 0$$

Notice in this case first basic feasible solution is possible by setting all non artificial variables to 0, we get positive feasible values of artificial variables.

# Simplex Method: Problem 2 contd..

Notice the contribution of slack variables and artificial variables to objective function  $z$  is 0.

This problem will be solved in two phases, in the first phase we will get first basic feasible solution without artificial variable.

To do that, we will maximize:  $-a_1 - a_2$

s.t.

$$2x_1 + x_2 - s_1 + a_1 = 5$$

$$x_1 + 3x_2 - s_2 + a_2 = 12$$

$$x_1, x_2, s_1, s_2, a_1, a_2 \geq 0$$

# Simplex Method: Problem 2, phase 1

|       |                | C <sub>j</sub>                  | 0              | 0              | 0              | 0              | -1             | -1   |      |        |
|-------|----------------|---------------------------------|----------------|----------------|----------------|----------------|----------------|------|------|--------|
|       |                | X <sub>1</sub>                  | X <sub>2</sub> | s <sub>1</sub> | s <sub>2</sub> | a <sub>1</sub> | a <sub>2</sub> | RHS  | L    |        |
| -1    | a <sub>1</sub> |                                 | 2              | 1              | -1             | 0              | 1              | 0    | 8    | 8      |
| -1    | a <sub>2</sub> |                                 | 1              | 3              | 0              | -1             | 0              | 1    | 12   | 4 →    |
|       |                | Z <sub>j</sub>                  | -3             | -4             | 1              | 1              | -1             | -1   | -20  |        |
|       |                | C <sub>j</sub> - Z <sub>j</sub> | 3              | 4              | -1             | -1             | 0              | 0    |      |        |
| <hr/> |                |                                 |                |                |                |                |                |      |      |        |
| -1    | a <sub>1</sub> |                                 | 5/3            | 0              | -1             | 1/3            | 1              | -1/3 | 4    | 12/5 → |
| 0     | X <sub>2</sub> |                                 | 1/3            | 1              | 0              | -1/3           | 0              | 1/3  | 4    | 12     |
|       |                | Z <sub>j</sub>                  | -5/3           | 0              | 1              | -1/3           | -1             | 1/3  |      |        |
|       |                | C <sub>j</sub> - Z <sub>j</sub> | 5/3            | 0              | -1             | 1/3            | 0              | -4/3 |      |        |
| <hr/> |                |                                 |                |                |                |                |                |      |      |        |
| 0     | X <sub>1</sub> |                                 | 1              | 0              | -3/5           | 1/5            | 3/5            | -1/5 | 12/5 |        |
| 0     | X <sub>2</sub> |                                 | 0              | 1              | 1/5            | -2/5           | -1/5           | -4/5 | 16/5 |        |

Entering variables, leaving variables are shown with arrows and pivot elements are showing with circles. We performed the same steps as the last problem.

Notice now we have a basic solution without artificial variable. Therefore, we import this solution to the next phase.

# Simplex Method: Problem 2, phase 2.

We will use this basic solution in the original objective function (the one with  $-z$ ). We import the basic solution by dropping columns corresponding to artificial variables and continue

| $C_j$ |             | -3       | -4       | 0           | 0           |             |   |
|-------|-------------|----------|----------|-------------|-------------|-------------|---|
|       |             | $X_1$    | $X_2$    | $s_1$       | $s_2$       | RHS         | L |
| -3    | $X_1$       | <b>1</b> | <b>0</b> | <b>-3/5</b> | <b>1/5</b>  | <b>12/5</b> |   |
| -4    | $X_2$       | <b>0</b> | <b>1</b> | <b>1/5</b>  | <b>-2/5</b> | <b>16/5</b> |   |
|       | $Z_j$       | -3       | -4       | 1           | 1           | -20         |   |
|       | $C_j - Z_j$ | 0        | 0        | -1          | -1          |             |   |



Notice all the  $C_j - Z_j$  values are negative or zero, therefore, we are done. -20 is the maximum value of  $-z$  therefore, the minimum value of  $z$  is 20 for  $X_1 = 12/5$  and  $X_2 = 16/5$ .